

```
ccrutter@csee-gpu01:~/FinalProj$ make
nvcc -arch=sm_75 -std=c++17 -O2 main.cu kernel.cu cpumergesort.cpp -o project
ccrutter@csee-gpu01:~/FinalProj$
```

Figure 1: Makefile

```
ccrutter@csee-gpu01:~/FinalProj$ ./project 1048576 256
Running FULL Merge Sort with N = 1048576, Block Size = 256
-----
Saved unsorted array to unsorted.txt
Saved CPU sorted array to cpu_sorted.txt
Saved GPU sorted array to gpu_sorted.txt
Appended timing results to sort_comparison.txt
ccrutter@csee-gpu01:~/FinalProj$
```

Figure 2: Sample Run

```
FinalProj > ≡ sort_comparison.txt
1  Array Size (N): 1048576 | Block Size: 256
2  CPU Sequential Merge Sort Time: 54.486 ms
3  GPU Parallel Merge Sort Time: 2.67965 ms
4  Speedup: 20.3333x
5  -----
6
```

Figure 3: Output

```

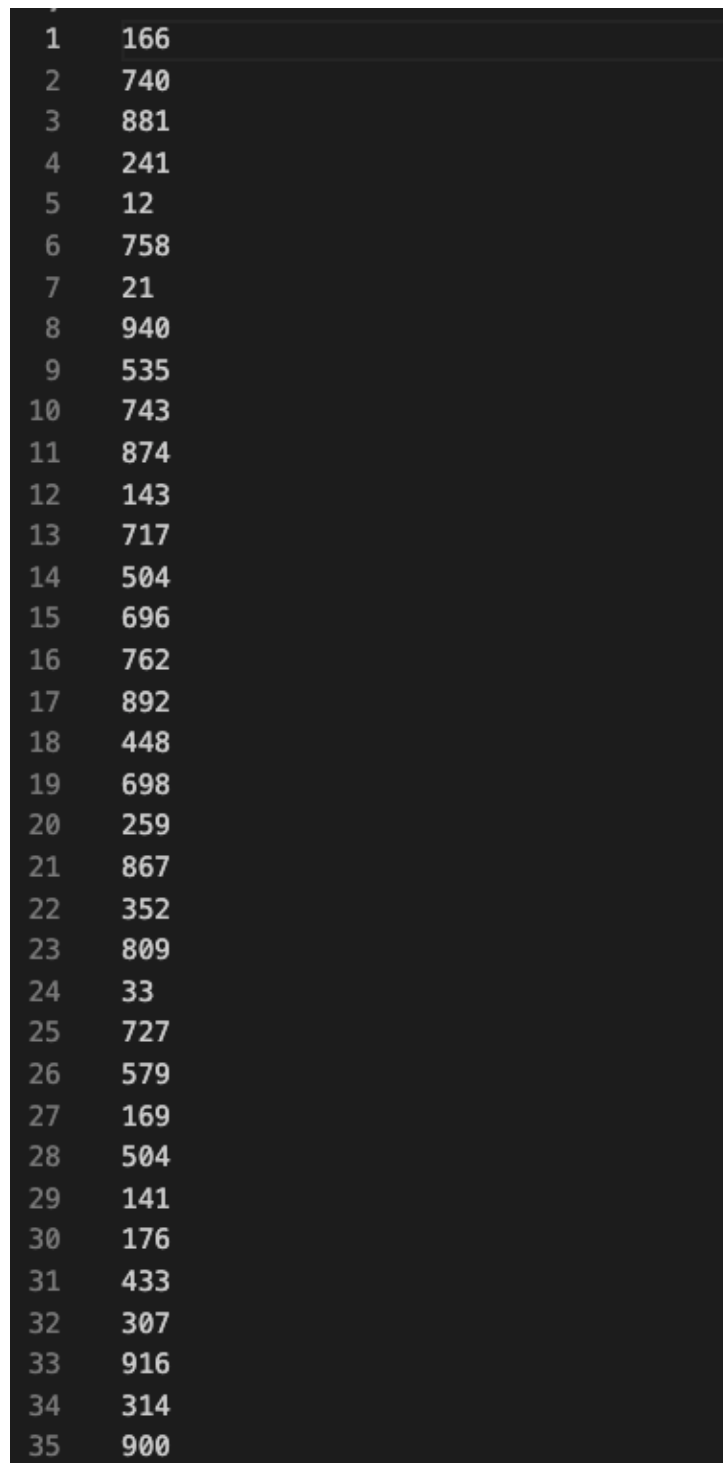
ccrutter@csee-gpu01:~/FinalProj$ ./run_tests.sh
Testing N=1024 with BlockSize=128
Running FULL Merge Sort with N = 1024, Block Size = 128
-----
Saved unsorted array to unsorted.txt
Saved CPU sorted array to cpu_sorted.txt
Saved GPU sorted array to gpu_sorted.txt
Appended timing results to sort_comparison.txt
Testing N=65536 with BlockSize=128
Running FULL Merge Sort with N = 65536, Block Size = 128
-----
Saved unsorted array to unsorted.txt
Saved CPU sorted array to cpu_sorted.txt
Saved GPU sorted array to gpu_sorted.txt
Appended timing results to sort_comparison.txt
Testing N=1048576 with BlockSize=128
Running FULL Merge Sort with N = 1048576, Block Size = 128
-----
Saved unsorted array to unsorted.txt
Saved CPU sorted array to cpu_sorted.txt
Saved GPU sorted array to gpu_sorted.txt
Appended timing results to sort_comparison.txt
Testing N=1024 with BlockSize=256
Running FULL Merge Sort with N = 1024, Block Size = 256
-----
Saved unsorted array to unsorted.txt
Saved CPU sorted array to cpu_sorted.txt
Saved GPU sorted array to gpu_sorted.txt
Appended timing results to sort_comparison.txt
Testing N=65536 with BlockSize=256
Running FULL Merge Sort with N = 65536, Block Size = 256
-----
Saved unsorted array to unsorted.txt
Saved CPU sorted array to cpu_sorted.txt
Saved GPU sorted array to gpu_sorted.txt
Appended timing results to sort_comparison.txt
Testing N=1048576 with BlockSize=256
Running FULL Merge Sort with N = 1048576, Block Size = 256
-----
Saved unsorted array to unsorted.txt
Saved CPU sorted array to cpu_sorted.txt
Saved GPU sorted array to gpu_sorted.txt
Appended timing results to sort_comparison.txt
Testing N=1024 with BlockSize=512
Running FULL Merge Sort with N = 1024, Block Size = 512
-----
Saved unsorted array to unsorted.txt
Saved CPU sorted array to cpu_sorted.txt
Saved GPU sorted array to gpu_sorted.txt
Appended timing results to sort_comparison.txt
Testing N=65536 with BlockSize=512
Running FULL Merge Sort with N = 65536, Block Size = 512

```

Figure 4: Bash Test File Run

```
Array Size (N): 1048576 | Block Size: 256
CPU Sequential Merge Sort Time: 54.486 ms
GPU Parallel Merge Sort Time: 2.67965 ms
Speedup: 20.3333x
-----
Array Size (N): 1024 | Block Size: 128
CPU Sequential Merge Sort Time: 0.042334 ms
GPU Parallel Merge Sort Time: 0.031584 ms
Speedup: 1.34036x
-----
Array Size (N): 65536 | Block Size: 128
CPU Sequential Merge Sort Time: 3.33813 ms
GPU Parallel Merge Sort Time: 0.114208 ms
Speedup: 29.2285x
-----
Array Size (N): 1048576 | Block Size: 128
CPU Sequential Merge Sort Time: 56.0159 ms
GPU Parallel Merge Sort Time: 1.65693 ms
Speedup: 33.8071x
-----
Array Size (N): 1024 | Block Size: 256
CPU Sequential Merge Sort Time: 0.043259 ms
GPU Parallel Merge Sort Time: 0.048128 ms
Speedup: 0.898832x
-----
Array Size (N): 65536 | Block Size: 256
CPU Sequential Merge Sort Time: 3.36797 ms
GPU Parallel Merge Sort Time: 0.172032 ms
Speedup: 19.5776x
-----
Array Size (N): 1048576 | Block Size: 256
CPU Sequential Merge Sort Time: 55.6716 ms
GPU Parallel Merge Sort Time: 2.49078 ms
Speedup: 22.3511x
-----
```

Figure 5: Bash Results



1	166
2	740
3	881
4	241
5	12
6	758
7	21
8	940
9	535
10	743
11	874
12	143
13	717
14	504
15	696
16	762
17	892
18	448
19	698
20	259
21	867
22	352
23	809
24	33
25	727
26	579
27	169
28	504
29	141
30	176
31	433
32	307
33	916
34	314
35	900

Figure 6: Unsorted.txt

```
FinalProj > ≡ cpu_sorted.txt
 1    0
 2    2
 3    2
 4    2|
 5    3
 6    3
 7    3
 8    4
 9    5
10    5
11    6
12    7
13    8
14    8
15   12
16   12
17   13
18   13
19   16
20   16
21   18
22   18
23   19
24   19
25   21
26   21
27   21
28   21
29   22
30   22
31   23
32   24
33   26
34   26
35   27
36   28
```

Figure 7: cpu_sorted.txt with 1024, and a 256 blocksize

FinalProj >	≡ gpu_sorted.txt
1	0
2	2
3	2
4	2
5	3
6	3
7	3
8	4
9	5
10	5
11	6
12	7
13	8
14	8
15	12
16	12
17	13
18	13
19	16
20	16
21	18
22	18
23	19
24	19
25	21
26	21
27	21
28	21
29	22
30	22
31	23
32	24
33	26
34	26
35	27
36	28

Figure 8: gpu_sorted.txt with same input size and blocksize